

Regression and correlation

• Regression is the study of dependency.

• It is used to answer:

(a) Does changing the class size affect the success of student?

(b) Can we predict height of a daughter from the height of mother?

• For this firstly, we need data.

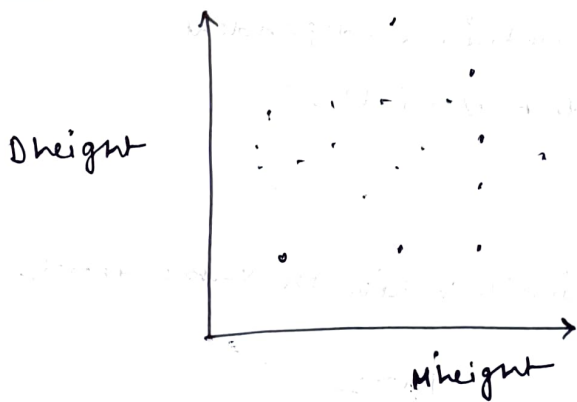
• Then, start with one predictor variable (X) and one response variable (Y)

Data consist of values (x_i, y_i) $i = 1, 2, \dots, n$ of (X, Y) observed on each of n unit cases.

Example! Inheritance of height

The collection of mother's height (M height) and corresponding daughter's height (D height) - inch

scatter plot



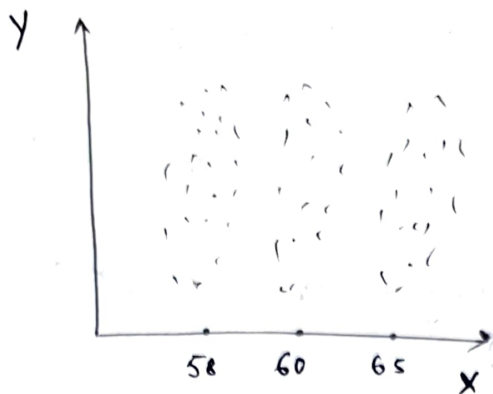
Answer

Do taller mother tend to have taller daughter?

Do shorter mother tend to have shorter daughter?

→ more understandable plot.

Scatter plot showing only pairs where mother's height round to 58, 64, 68 inch



- Here, we may notice some trends in data.
- One interest how the distribution of y changes as x is varied.
- One important aspect of distribution is mean function and variance function.

Correlation

A correlation describe a relationship between two variables.

Correlation: is there a relationship between two variables?

Regression: how well a certain independent variable predict dependent variable?

Type of correlation

Positive: If both the variable varies in same direction.

Height	158	160	...	172
Weight	52	55	60	

Negative Correlation

price of a product	6	5	4	3	2	1
demand	72	120	175	250	260	-

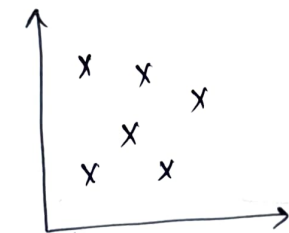
Zero correlation

If we don't have any relation between the two variables.

height $\rightarrow$ intelligence

Measurement of correlation

① Scatter plot



Zero correlation

Variance

Give information on variability of a single variable

$$S_x^2 = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n-1}$$

* $(n-1)$ due to Bessel correction is the adjustment we make when calculating the sample variance instead of population variance.

Co-variance

$$\text{Cov}(x, y) = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{n-1}$$

Calculation of co-variance

1.

x	y
2	20
3	60
6	80

$$\text{Cov}(x, y) = 56.67$$

2.

x	y
2	30
4	60
6	80
8	100

$$\text{Cov}(x, y) = ?$$

3.

x	y
1	4
2	5
3	4

$$\text{Cov}(x, y) = ?$$

Problem in covariance

- dependent on size of the data's standard deviation
- If large, value is large, if small, then value is small.

X	Y
101	100
81	80
61	60
51	50
41	40
21	20
1	0
51	50

covariance = 1166.67

x	y
54	53
53	52
52	51
51	50
50	49
49	48
48	47
51	50

covariance = 4.67

Solution / Pearson's r (Pearson correlation coefficient)

- Covariance does not really tell us anything
Solution! standardise this measure
- Pearson's r: standardise the covariance
- Divide the covariance by the multiplied standard deviations of x and y.

$$r_{xy} = \frac{\text{cov}(x, y)}{s_x s_y}$$

$$r_{xy} = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2} \sqrt{\sum (y - \bar{y})^2}}$$

Exercise:

- ① Find Pearson correlation coefficient between advertisement expenses and sales volume.
- Advertisement Exp (in Lakh): 11, 13, 14, 16, 16, 15, 15, 14, 13, 13
- Sales Volume (in Lakh): 50, 50, 55, 60, 65, 65, 65, 60, 60, 50

2. Find the correlation coefficient between age and playing habits of the following students using Karl Pearson.

Age	15	16	17	18	19	20
no of student	250	200	150	120	100	80
Regular Player	200	150	90	48	30	12

Percentage of playing habit

$$= \frac{\text{No. of regular player}}{\text{Total no. of student}} \times 100$$

Pearson's (measure linear relationship between x and y)

$$\text{range} - \leq r \leq 1$$

- $r=1$ Perfect positive linear correlation
- $r=-1$ Perfect negative linear correlation
- $r=0$ NO linear correlation.

→ still there may be non-linear one

→ close to $(+1)$ or (-1) strong linear relationship

→ close to (0) weak or no linear relationship

Exercise

Calculate Pearson's r where $y = x^2$ and x ranges from -3 to $+3$?

$$x = [-3, -2, -1, 0, 1, 2, 3]$$

Note: There are some correlation coefficient for non-linear relationship.

— Spearman's rank correlation coefficient

Recap one analysis of data

$$\bar{x} = \text{sample average of } x = \frac{\sum x_i}{n}$$

$$\bar{y} = \text{sample average of } y = \frac{\sum y_i}{n}$$

$$s_x = \text{sample standard deviation}$$

$$s_x^2 = \text{sample variance of } x = \frac{\sum (x_i - \bar{x})^2}{n-1}$$

$$s_y = \text{sample standard deviation}$$

$$s_y^2 = \text{sample variance of } y = \frac{\sum (y_i - \bar{y})^2}{n-1}$$

$$\text{cov}(x, y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n-1} = \text{sample covariance}$$

$$r_{xy} = \frac{\text{cov}(x, y)}{s_x s_y} = \text{sample correlation}$$